

GTFS2GMNS: Reading & Concept Checklist

Ensuring You Understand Before You Compute

Core Principle

GTFS files are not static data—they are decision variables.

`stops.txt` → Where should we place stops? (coverage vs. speed)

`stop_times.txt` → How often should buses run? (frequency vs. fleet cost)

`routes.txt` → Which corridors should we serve? (demand vs. equity)

1 Reading 1: Valley Metro Service Planning (Lecture 2.1)

Document: Joe Gregory's Presentation (27 slides)

Who: Joe Gregory, Manager of Service Planning & GIS, Valley Metro

Focus: Real-world transit planning in Phoenix metropolitan area

Reading Checklist

- ☐ Read slides 1–7: System perspectives (Riders, Operators, Funders)
 - ☐ Read slides 8–12: Origin-Destination Study methodology
 - ☐ Read slides 13–16: **Fiesta BUZZ Case Study** (critical!)
 - ☐ Read slides 17–21: Autonomous Vehicle Pilot integration
 - ☐ Read slides 22–27: Human Spectrogram discussion questions

Concept Verification Questions

Questions from Lecture 2.1 (Fill in or answer briefly)

Q1. The Valley Metro O-D study collects _____ completed surveys every 3–5 years.

Q2. In the Fiesta BUZZ case (slide 14), the round-trip runtime was _____ minutes, and On-Time Performance suffered because drivers lacked sufficient _____ time.

Q3. The solution (slide 16) involved three changes: (a) _____ routes, (b) add _____ vehicle(s), and (c) adjust _____ locations.

Q4. Why does “interlining” (connecting DBUZ and FBUS) help reduce fleet requirements?

Q5. From slide 25, should transit be run more like a social service or a business? What's your view and why? (2 sentences)

2 Reading 2: Bus and Driver Scheduling (Lecture 2.2)

Document: Scheduling Overview (PowerPoint)

Focus: The mathematics connecting timetables to fleet size and crew requirements

Reading Checklist

- ☐ Understand the definition of **cycle time** (round-trip including layovers)
 - ☐ Understand the definition of **deadhead** (non-revenue vehicle movement)
 - ☐ Understand the definition of **blocking** (assigning trips to vehicles)
 - ☐ Memorize the fleet equation: $N = T_{\text{cycle}}/h$

Concept Verification Questions

Questions from Lecture 2.2 (Calculate or explain)

Q6. Write the fleet equation: $N_{\text{vehicles}} = \underline{\hspace{2cm}}$

Q7. If cycle time = 60 min and headway = 10 min, how many vehicles are needed? $\underline{\hspace{2cm}}$

Q8. If you want to **double** frequency (halve headway), what happens to fleet size? $\underline{\hspace{2cm}}$

Q9. Define “deadhead” in one sentence:

Q10. Why is minimizing deadhead important for transit agencies? (cost perspective)

3 Reading 3: GTFS2GMNS User Guide (Lecture 2.4)

Document: Technical User Guide (15 pages)

Source: ASU Trans+AI Lab

Focus: Converting GTFS to network format for modeling and optimization

Reading Checklist

- ☐ Read Section 1: Overview (what GTFS2GMNS does)
 - ☐ Read Section 2: Data flow structure (Figure 1 is essential!)
 - ☐ Read Section 4: Illustrative Example (two-layer network)
 - ☐ Read Section 5: BART case study
 - ☐ Read Section 6.1–6.4: Applications (Assignment, Accessibility, Equity)
 - ☐ Read Section 6.5: Linear Programming formulations

Concept Verification Questions

Questions from Lecture 2.4 (Identify and explain)

Q11. GTFS2GMNS produces two output files: _____ and _____

Q12. List the **four types of nodes**:

1. Physical nodes (stops/stations)
2. _____ nodes

Q13. List the **four types of links**:

1. Service links (connect stops along routes)
2. _____ links (passenger boards)
3. _____ links (passenger alights)
4. _____ links (walk between stops)

Q14. In the BART example (Section 5.3.1), how many physical stations? _____ Routes? _____

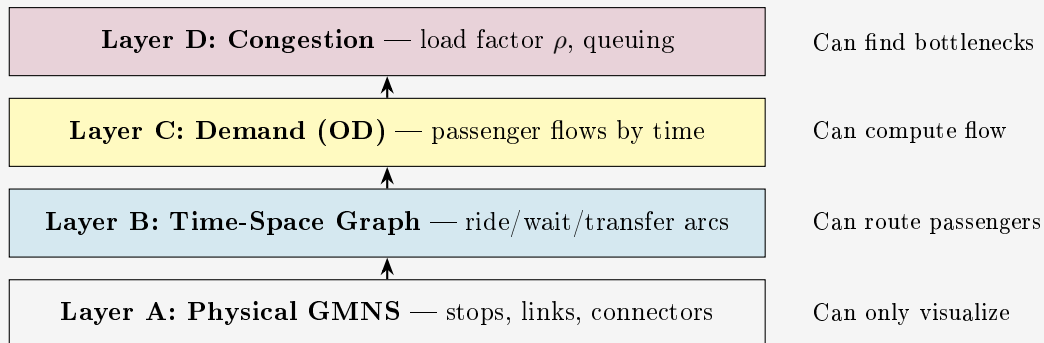
Q15. What field in `link.csv` contains travel time? _____

Q16. What field in `link.csv` contains vehicle/link capacity? _____

Q17. From Section 6.5, the transit assignment LP minimizes total _____ subject to flow conservation and _____ constraints.

4 The Four-Layer Framework

Understanding the Layers



Layer Understanding Check

Q18. If you only have Layer A (`node.csv` + `link.csv`), can you answer “how long does it take to travel from stop X to stop Y at 8:00 AM?” Yes / No

Q19. Which layer do you need to answer “which route segment is overcrowded?” _____

Q20. The load factor ρ is defined as: $\rho = \frac{\text{_____}}{\text{_____}}$

5 Key Formulas Reference

Formulas You Should Know

1. Circuitry Index (route directness):

$$\text{Circuitry} = \frac{\text{Route Length}}{\text{Straight-line Distance}}$$

Interpretation: ≈ 1 = express; > 2 = very circuitous

2. Headway Coefficient of Variation (service regularity):

$$CV_h = \frac{\sigma_h}{\bar{h}}$$

Interpretation: < 0.2 = regular; > 0.5 = severe bunching

3. Fleet Size Equation:

$$N_{\text{vehicles}} = \frac{T_{\text{cycle}}}{h} = \frac{2 \times T_{\text{run}} + 2 \times T_{\text{layover}}}{h}$$

4. Load Factor:

$$\rho = \frac{\text{Passengers on link}}{\text{Capacity}}$$

Action required if $\rho > 1.0$

5. Effective Speed:

$$v_{\text{eff}} = \frac{\text{Distance}}{\text{Total Time (including dwell)}}$$

If < 10 mph, consider stop consolidation

6 Connection to Project Applications

How This Connects to Your Project

Your project will use GTFS2GMNS output for one of these analysis tracks:

Group 1: QGIS & Land Use Analysis

- Number of stops per zone
- Average walking distance from stops to zone centers
- Zones with highest demand
- POI (Points of Interest) density per zone

Group 2: Frequency & Equity Analysis

- Number of routes per zone
- Frequency count per zone
- Zones accessible within 45-minute transit time
- Stops-to-demand ratio (equity metric)

Group 3: Operations & Scheduling Analysis

- Select a major line
- Count stops, calculate frequency by time-of-day
- Estimate vehicles and drivers needed
- Create Gantt chart showing vehicle blocks

Final Integration: DTALite Runs

- `od_accessibility.csv` — accessibility metrics
- `routes.csv` — paths with transfer counts
- Convergence analysis across iterations

Project Data Sources**Data You Will Use**

- ☐ **GTFS feeds** from your assigned city
- ☐ **Road networks:** Chicago, Phoenix, Atlanta, Philadelphia, Anaheim, Barcelona, Berlin-Center, Birmingham-England, Eastern-Massachusetts, Munich
- ☐ **Sources:**
 - <https://github.com/bstabler/TransportationNetworks>
 - OpenStreetMap via `osm2gmns`
- ☐ **Zone data:** Traffic Analysis Zones (TAZ) with demographics

7 Self-Assessment: Am I Ready?**Readiness Checklist (Check all that apply)****Conceptual Understanding:**

- I understand why `stops.txt` is a “decision variable”
- I can explain the difference between physical nodes and service nodes
- I can explain what the four link types represent
- I understand why Layer A alone is not enough for analysis
- I can calculate fleet size given cycle time and headway

Practical Skills:

- I know how to run GTFS2GMNS (set input/output paths, time period)
- I can interpret `node.csv` fields
- I can interpret `link.csv` fields
- I know where to find GTFS data for my city

Connection to Applications:

- I understand how GMNS output feeds into traffic assignment
- I understand how to calculate accessibility metrics
- I can connect the Fiesta BUZZ case to my own analysis

8 Submission**What to Submit:**

1. This checklist with all boxes checked and questions answered
2. Brief confirmation that you successfully ran GTFS2GMNS on a test dataset

Grading: Complete/Incomplete based on evidence of reading engagement

Student Name: _____

Date: _____